Interest Calculator

A menu-driven Python application for interest, loan and investment calculations

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What it does

Simple interest

Interest and total amount for any principal, rate and time.

Compound interest

Yearly, half-yearly, quarterly, monthly or daily compounding.

Simple vs compound

A year-by-year table showing how the two diverge.

Loan EMI

Monthly EMI, total interest and a full amortization schedule.

SIP calculator

Future value of a fixed monthly investment.

Time to double

Rule of 72 estimate next to the exact answer.

History

Every calculation of the session, listed on demand.

CSV export

Save the history to a file that opens in Excel.

Formulas used

Calculation	Formula
Simple interest	SI = P x R x T / 100
Compound amount	A = P x (1 + R / (100n)) ^ (nT)
Loan EMI	P x r x (1+r)^N / ((1+r)^N - 1)
SIP future value	M x (((1+i)^N - 1) / i) x (1+i)
Doubling time	72 / R and ln 2 / ln(1 + R/100)

How the code is organised

Formula functions

Plain functions such as simple_interest and calculate_emi. They only take numbers and return numbers, so they are easy to test.

Input helpers

ask_number repeats the question until the user types a valid value. Letters, negatives and zero tenure are rejected.

Menu screens

Each option is a do_ function. A loop in main shows the menu, runs the choice and stores results in a history list.

Sample output

Compound interest

```
Principal: Rs. 10,000.00
Rate: 10% Time: 2 years (Monthly)
Interest: Rs. 2,203.91
Total amount: Rs. 12,203.91
```

Loan EMI

```
Loan amount: Rs. 100,000.00
Rate: 10% Tenure: 6 months
Monthly EMI: Rs. 17,156.14
Total interest: Rs. 2,936.84
```

Skills and next steps

Skills used

Functions, loops and conditions, try/except validation, lists and dictionaries, string formatting, CSV file handling, the math and datetime modules.

Next steps

A Tkinter interface, charts with Matplotlib, history saved between sessions, and a tax or GST calculator.

Thank you. Questions?